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FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Main)-2024

TEST DATE: 22-03-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

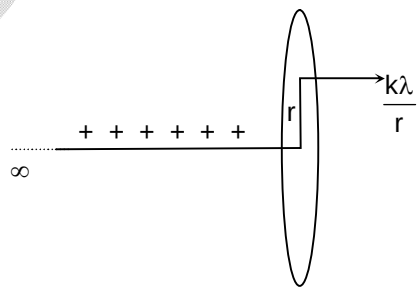
1. B

Sol. $\int_0^R \frac{k\lambda}{r} 2\pi r dr = \frac{q}{4\varepsilon_0}$

$k\lambda 2\pi R = \frac{q}{4\varepsilon_0}$

$\frac{\lambda R}{2\varepsilon_0} = \frac{q}{4\varepsilon_0}$

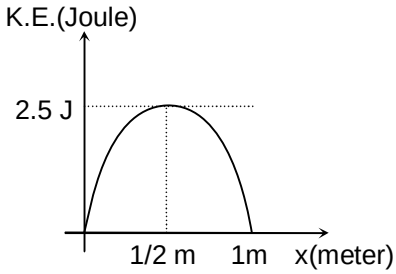
$\Rightarrow \quad \lambda = \frac{q}{2R}$



2. B

Sol. $F - \mu mg = mv\frac{dv}{dx}$

$10x - 10x^2 = K.E.$

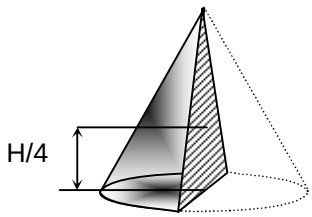


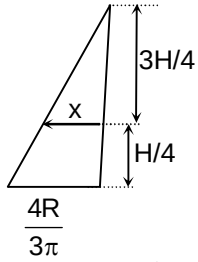
3. C

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Sol. $\frac{4x}{3H} = \frac{4R}{3\pi H}$

$\Rightarrow x = \frac{R}{\pi}$





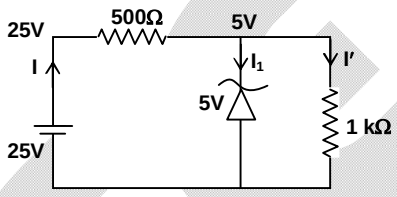
4. C

Sol. $I = \frac{20}{500} A$

$I' = \frac{5}{1000} = \frac{1}{200} A$

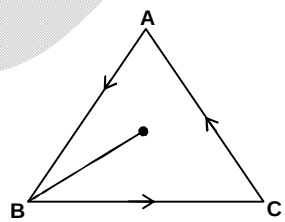
$I_1 = \frac{1}{25} - \frac{1}{200} = \frac{7}{200}$

$\frac{I}{I_1} = \frac{1/25}{7/200} = \frac{8}{7}$



5. C

Sol. Average velocity = $\frac{\ell/\sqrt{3}}{\text{time}} = \frac{\ell/\sqrt{3}}{2\ell/3v} = \frac{\sqrt{3}v}{2}$



6. C

Sol. $I_R = I_T - I_{cut} = \frac{(\sigma\pi R^2)R^2}{4} - \left[\frac{\left(\frac{\sigma\pi R^2}{4}\right)\frac{R^2}{4}}{4} + \left(\frac{\sigma\pi R^2}{4}\right)\frac{R^2}{8}\right]$

$= \frac{\sigma\pi R^4}{4}\left[1 - \frac{1}{16} - \frac{1}{8}\right] == \frac{13\sigma\pi R^4}{64}$

7. A

Sol. $P = \frac{E_{rms}^2 \cos\phi}{z} \Rightarrow 400 = \frac{(250)^2}{z} \times 0.8$

$\cos\phi = \frac{R}{Z} = 0.8$

$z = \sqrt{X_L^2 + R^2}$

$R = 100\ \Omega$

$Z = 125\ \Omega$

$X_L = 75\ \Omega$

8. B

Sol. $m_1v_1 = m_2v_2$

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$m_1 = m_2 8$

$\frac{m_1}{m_2} \Rightarrow \frac{R_1}{R_2} = (8)^{1/3} = 2:1$

9. B

Sol. $\frac{dQ}{dt} = \frac{k_1 40A}{L} = \frac{k_2 10A}{L} = \frac{k_3 50A}{L}$

Under steady state $\Rightarrow 4k_1 = k_2 = 5k_3$

10. C

Sol. induced emf $e = (-)L\left(\frac{dI}{dt}\right)$

11. C

Sol. Gravitational potential at point P

$V_P = -\frac{GM}{R}$

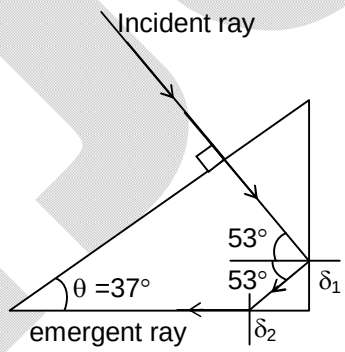
Work done, $W = m_0\left[V_\infty - V_P\right] = \frac{GMm_0}{R}$

12. C

Sol. $\delta_1 = 180° - 2 \times 53° = 74°$

$\delta_2 = 53°$

$\delta_{net} = \delta_1 + \delta_2 = 127°$



13. B

Sol. $15 \times \frac{2\pi m}{Bq} v\cos 37° = d$

14. C

Sol. No. of photons per second = $\frac{I}{\left(\frac{hC}{\lambda}\right)} \times \text{Area} = \frac{8 \times 10 \times 10^{-4}}{2 \times 1.6 \times 10^{-19}}$

No. of photoelectrons per second = $\frac{1}{5} \times \frac{8 \times 10 \times 10^{-4}}{2 \times 1.6 \times 10^{-19}} = 5 \times 10^{15}$

15. A

Sol. $\left[\beta x^2\right] = \left[k\theta\right]$

$\beta L^2 = ML^2T^{-2}$

$\beta = MT^{-2}$

Also,

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$[W] = [\alpha\beta]$

$\left[ML^2T^{-2}\right] = \left[\alpha MT^{-2}\right]$

$\alpha = L^2$

16. B

Sol. $P_0V_0^2 = P4V_0^2$

$P = \frac{P_0}{4}$

$P_0V_0 = nRT_1$

$\frac{P_0}{4}(2V_0) = nRT_2$

$\Delta U = \frac{3}{2}nR(T_2 - T_1) = \frac{3}{2}(-)\frac{P_0V_0}{2} = -\frac{3}{4}P_0V_0$

17. B

Sol.

A	B	Y
1	0	1
1	1	1
0	0	0
0	1	1

18. C

Sol. $T_1 = 2\pi\sqrt{\frac{\ell}{g}},\ T_2 = 2\pi\sqrt{\frac{\ell}{g+a}}$

$T_2 = 2\pi\sqrt{\frac{\ell}{(g+4)}}$

$y = 2t^2,\ a = \frac{d^2y}{dt^2} = 4\ m/s^2$

$\left(\frac{T_2}{T_1}\right)^2 = \left(\frac{g}{g+4}\right) = \frac{10}{14} = \frac{5}{7}$

19. B

Sol. Potential energy

$U_i = -\vec{P}_2 \cdot \vec{E}_1 = -P\left(\frac{2kP}{d^3}\right)\cos 180°$

$U_i = \frac{2kP^2}{d^3} = \frac{P^2}{2\pi\varepsilon_0 d^3}$

$U_f = \frac{P^2}{16\pi\varepsilon_0 d^3}$

$\frac{P^2}{2\pi\varepsilon_0 d^3} - \frac{P^2}{16\pi\varepsilon_0 d^3} = \frac{1}{2}mv^2 \times 2$

$v = \sqrt{\frac{7P^2}{16\pi m\varepsilon_0 d^3}} = \frac{P}{4d}\sqrt{\frac{7}{\pi m\varepsilon_0 d}}$

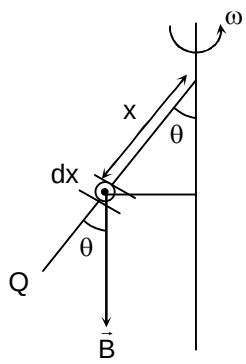
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20. B

Sol. $de = (B \sin\theta)\, dx \times \sin\theta\, \omega$

$$e = B\omega \sin^2\theta \int_0^L x\,dx$$

$$e = \frac{1}{2} B\omega L^2 \sin^2\theta$$



SECTION – B

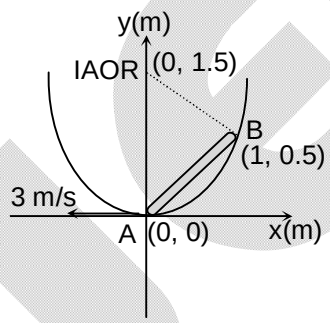
21. 2

Sol. IAOR at (0, 1.5)

$v = r\omega$

$3 = 1.5\omega$

$\omega = 2$ rad/s



22. 2

Sol. $$\phi = \frac{2\pi}{\lambda} = \frac{2\pi}{\lambda}\frac{\lambda}{4}$$

$$\phi = \frac{\pi}{2}$$

$$I_R = I_0 \cos^2\left(\frac{\phi}{2}\right) = I_0 \cos^2\frac{\pi}{4}$$

$$I_R = \frac{I_0}{2}$$

23. 7

Sol. $$Av_1 = \frac{A}{4} v_2 \Rightarrow v_2 = 4v_1$$

$$P_1 + \rho gh + \frac{1}{2}Pv_1^2 + P_2 + \frac{1}{2}Pv_2^2$$

$$(P_1 - P_2) + \rho gh = \frac{1}{2}P\left(v_2^2 - \frac{v_2^2}{16}\right)$$

$$(P_1 - P_2) + \rho gh = \frac{15}{32}\rho v_2^2$$

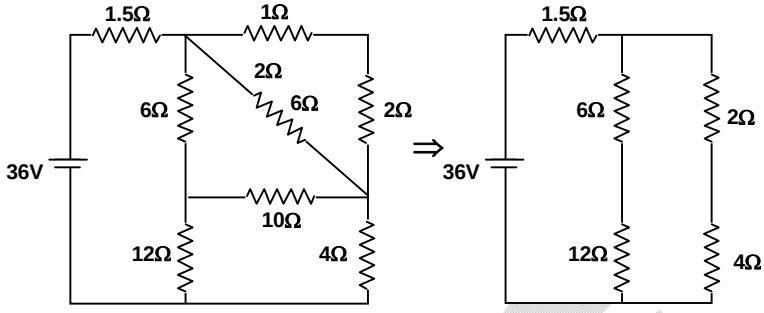
$\Rightarrow v_2 = 7$ m/s

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24. 6

Sol. $R_{eq} = 6\Omega \Rightarrow \dfrac{V}{R_{eq}} = I$

$I = 6A$



25. 4

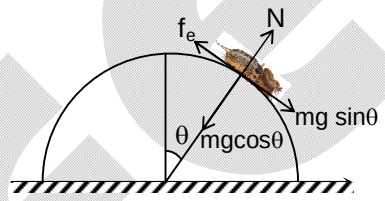
Sol. At slipping

$\tan\theta = \mu_S$

$\tan\theta = \dfrac{1}{\sqrt{3}}$

$\theta = \dfrac{\pi}{6}$

$\ell = R\theta = \dfrac{24}{\pi} \times \dfrac{\pi}{6} = 4$ cm



26. 885

Sol. $I = \dfrac{1}{2}\varepsilon_0 E_0^2 c$

Radiation pressure $= \dfrac{2I}{c} = \varepsilon_0 E_0^2 = 8.85 \times 10^{-12} \times 10^4 = 885 \times 10^{-10}$

27. 20

Sol. $T_1 + T_2 = 20$

$T_2 L = 20\,x$

$T_2 = \dfrac{20x}{L} = 20x$

$T_1 = 20(1-x)$

Also,

$\dfrac{1}{2\ell}\sqrt{\dfrac{T_1}{\mu}} = \dfrac{1}{\ell}\sqrt{\dfrac{T_2}{\mu}}$

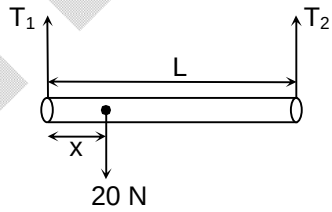
$T_1 = 4T_2$

$20(1-x) = 80\,x$

$20 - 20x = 80x$

$\Rightarrow \dfrac{20}{100} = x$

$\Rightarrow x = 20$ cm



28. 4

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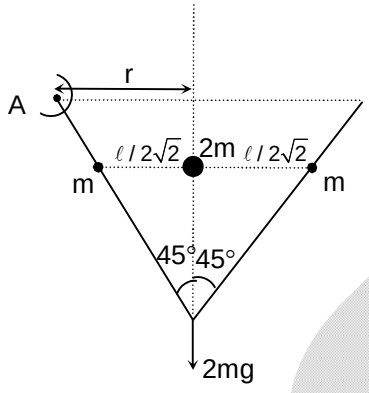
Sol. $\tau = 2mg \times \frac{\ell}{\sqrt{2}}$

$$\left[\frac{m\ell^2}{3} + \frac{m\ell^2}{12} + m\left(\frac{\sqrt{5}\ell}{2}\right)^2\right]\alpha = 2mg\frac{\ell}{\sqrt{2}}$$

$$\Rightarrow \frac{5}{3}\ell\alpha = \frac{2g}{\sqrt{2}}$$

$$\frac{5}{3}\frac{3}{\sqrt{2}}\alpha = \frac{20}{\sqrt{2}}$$

$\Rightarrow \alpha = 4$ rad/s^2



29. 4

Sol. $\frac{dy}{dx} = \tan\theta = 8x$

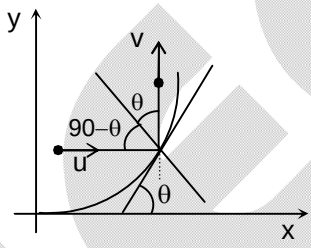
$u\cos\theta = v\sin\theta$

$eu\sin\theta = v\cos\theta$

$\tan^2\theta = 1/e$

$\tan\theta = 2$

$x = 1/4$ and $y = 1/4$



30. 4

Sol. $\text{Charge} = \frac{|\Delta\phi|}{R} = \frac{B\pi r^2}{4R}$

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Chemistry

PART – B

SECTION – A

31. C

Sol. Ionic product of $MOH = \frac{0.2}{2}\left[\frac{10^{-5}}{2}\right]$

$= \frac{0.1}{2} \times 10^{-6}$

$= 5 \times 10^{-7}\ M^2$

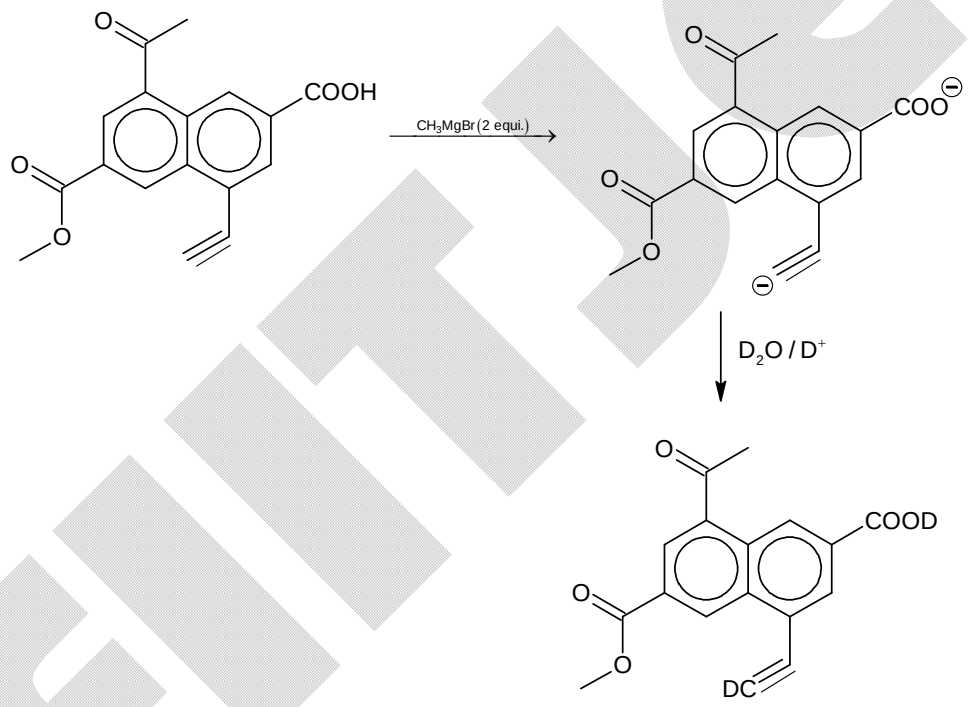
K_{ip} for $M(OH)_2 = \frac{0.2}{2}\left[\frac{10^{-5}}{2}\right]^2 = \frac{0.1}{4} \times 10^{-10}$

$= 2.5 \times 10^{-12}\ M^3$

Only for $Pb(OH)_2$, $K_{ip} > K_{sp}$, hence $Pb(OH)_2$ for precipitate.

32. D

Sol.



33. B

Sol. $L_1 = 179$ pm, $L_2 = 163$ pm

34. A

Sol. $\Delta W = \frac{\pi(5V_1 - V_1)(7P_1 - P_1)}{4} = \frac{\pi 4V_1 \times 6P_1}{4}$

$\Delta W = 6\pi P_1 V_1$

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35. A

Sol. $t_{1/2} = \frac{A_o}{2k} \Rightarrow k = \frac{A_o}{200}$

For zeroth $A_o - A = +kt$

$A_o - \frac{A_o}{8} = 8kt$

$\frac{7A_o}{8} = \frac{A_o}{200}t$

$t = \frac{700}{4} = 175$ min

36. D

Sol. $K_c = \frac{[D]^3}{[A][B]^2}$ unitless

$= M^o$

37. D

Sol. Due to relative lowering of vapour pressure.

38. B

Sol. pK_a, 2-fluorobenzoic acid = 3.27
2-bromobenzoic acid = 2.88
2-chlorobenzoic acid = 2.94
Due to SIR effect.

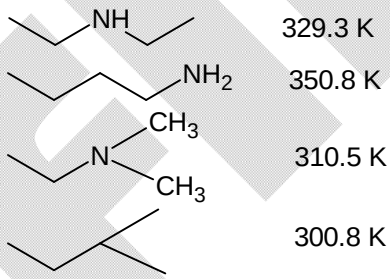
39. B

Sol. Vitamin B_1 – Beri-Beri
Rest are corrects.

40. A

Sol. Zn > Co > Cu > Ni
906 758 745 736 (kJ/mole)

41. C

Sol. 

329.3 K
350.8 K
310.5 K
300.8 K

42. D

Sol. $\Delta T_f \propto$ im

Higher value of im, lower the freezing point.

(i) im $= 2 \times 2 = 4$ (ii) im $= 5 \times 3 = 15$

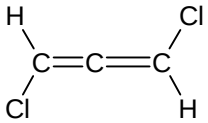
(iii) im $= 3 \times 4 = 12$ (iv) im $= 2 \times 5 = 10$

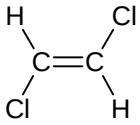
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43. A
Sol. Potassium has lilac colour on flame test.

44. C
Sol. $\left[Co(en)_2(NH_3)_2\right]^{+3}$ has 2 geometrical isomerism and it's cis form will have optical isomerism.

45. A
Sol. AgBr is a partially soluble in NH_3 and in $Na_2S_2O_3$, NaCN, Ag^+ form complexes.

46. B
Sol.  Non-planar structure

 Planar and trans forms

47. C
Sol. Maltose on hydrolysis give two molecular of glucose only.

48. B
Sol. $CrO_4^{-2} + H_2O_2 + H^+ \longrightarrow CrO_5 + 3H_2O$

$4CrO_5 + 12H^+ \longrightarrow \underset{\text{green soln}}{4Cr^{+3}} + 7O_2 + 6H_2O$

49. A
Sol. $M^{+3} \xrightarrow{xV} M^{+2} \xrightarrow{z} M^{+1} \xrightarrow{?} M$

$\underbrace{\qquad\qquad\qquad}_{y}$

$x \times 1 + z \times 1 + Q \times 1 = y \times 3$

$Q = 3y - x - z$

50. C
Sol. $E = \left(\frac{hc}{\lambda}\right)$, λ should be reverse order of strength of ligands.

SECTION – B

51. 6
Sol. Elements follow $4d^{1\text{ to }10}\ 5s^1$ are Nb, Mo, Tc, Ru, Rh, Ag.

52. 5
Sol. Mischmetal is alloy of lanthanoid, which contain nearly 5% of Fe.

53. 12
Sol. By the ratio find % of C, H and O then formula

$C_3H_6O + 4O_2 \longrightarrow 3CO_2 + 3H_2O$

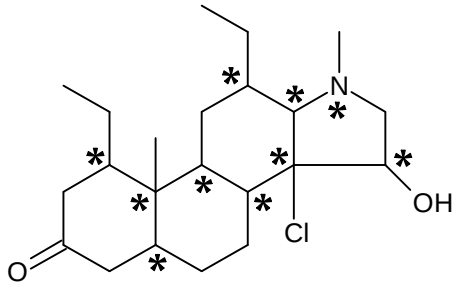
Y

3 mole compound will require 12 mole O_2.

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54. 10

Sol.



55. 675

Sol. Balmer series $\frac{1}{\lambda_1} = R_H Z^2 \left(\frac{1}{2^2} - \frac{1}{4^2}\right)$ .... (1)

Paschen series $\frac{1}{\lambda_2} = R_H Z^2 \left(\frac{1}{3^2} - \frac{1}{4^2}\right)$ .... (2)

$$\frac{\frac{1}{\lambda_1}}{\frac{1}{\lambda_2}} = \left(\frac{\frac{1}{2^2} - \frac{1}{4^2}}{\frac{1}{3^2} - \frac{1}{4^2}}\right) = \left(\frac{\frac{4-1}{16}}{\frac{16-9}{9\times16}}\right) = \frac{3}{16} \times \frac{9\times16}{7}$$

$$\frac{\lambda_2}{\lambda_1} = \frac{27}{7} \Rightarrow \frac{\lambda_2}{350} = \frac{27}{7}$$

$$\frac{\lambda_2}{175} = \frac{27}{7} \Rightarrow \lambda_2 = \frac{175\times27}{7}$$

$\lambda_2 = 675$ nm

56. 200

Sol. $M = \frac{20}{40} \times \frac{1000}{500} = 1\,M$

$M_1V = M_2V_2$

$1\times V_1 = 0.2\times1000$

$V_1 = 200$ ml

57. 18

Sol. $n = 5$

$\ell = 4$

g(sub shell)

$m = -4, -3, -2, -1, 0, +1, +2, +3, +4$

9 orbitals.

Maximum electron = 18.

58. 3

Sol. $NO(g) + \frac{1}{2}O_2(g) \rightleftharpoons NO_2(g)$

$\Delta G_r^o = \Delta G_{NO_2}^o - \Delta G_{NO}^o = 5.1454 - 1 = 4.1454$ Kcal

$\Delta G^o = -2.303RT \log_{10} K_{eq}$

$4.1454\times10^3 = -2.303RT \log_{10} K_{eq}$

$4.1454\times10^3 = -2.303\times2\times300 \log_{10} K_{eq}$

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$$\log_{10} K_{eq} = -\frac{4.1454}{2.303 \times 300 \times 2}$$

$$= -\frac{4145.4}{2.303 \times 600} = -3$$

$$K_{eq} = 10^{-3}$$

59. 70

Sol. x = 7, y = 10

60. 29

Sol. $E^{o}_{Cu^{2+}/Cu} = 0.34\ V$

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Mathematics

PART – C

SECTION – A

61. C

Sol. f(x) is discontinuous at $x = 1, \sqrt{2}, \sqrt{3}, 2, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{8}, 3$, g(x) is non differentiable at $x = 1, \sqrt{2}, \sqrt{3}, 2, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{8}$ and $\frac{3}{2}$.

62. B

Sol. $(x^2 - 4)(x - 1)(x^5 + 2x - 3) = 0$

$x^5 + 2x - 3 = 0$ has only real root.

63. B

Sol. $\begin{Bmatrix} x^2 - 12x + 37 & ; & x \le 2 \\ 2x - x^2 + 17 & ; & 2 < x < 5 \\ x^2 - 12x + 37 & ; & x \ge 5 \end{Bmatrix}$

64. C

Sol. $\lim\limits_{x \to 0} \dfrac{(A-B)x + \left(A - \frac{B}{2} + C\right)x^2 + \left(\frac{A}{2} - \frac{B}{3} - C\right)x^3 .....}{\frac{x^3}{2}} = 2024$

$\Rightarrow \quad A = B,\ A - \frac{B}{2} + C = 0, \frac{A}{2} - \frac{B}{3} - C = 1012$

$A = B = 1518,\ C = -759$

65. B

Sol. $J(x) = \int e^{\tan^2 x} \left(2\tan x\left(\sec^2 x \cos^2 x - \frac{1}{1+\tan^2 x}\right)\right)dx$

$= \int e^{\tan^2 x}(2\tan x \sec^2 x \cdot \cos^2 x - \sin 2x)dx$

$J(x) = e^{\tan^2 x}\cos^2 x + c$

$\because\ J(0) = 1 \ \Rightarrow c = 0$

$J\left(\frac{\pi}{4}\right) = \frac{e}{2}$

66. B

Sol. $\tan 45° = \left|\frac{m-2}{1+2m}\right| \Rightarrow m = -3,\ \frac{1}{3}$

$a\left|\frac{1}{m_1} - \frac{1}{m_2}\right| = 12.5$

$\Rightarrow \frac{10a}{3} = 12.5 \ \Rightarrow a = 3.75$

67. B

Sol. α and β lie on the chord $y = 1$ intersecting the circle $x^2 + y^2 - 8x - 6y + 17 = 0$

$\therefore \quad x^2 - 8x + 12 = 0 \ \Rightarrow x = 2, 6$

$\therefore \quad \alpha = 2 + i,\ \beta = 6 + i$

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68. D

Sol. $\tan 45° = \left|\frac{m-3}{1+3m}\right| \Rightarrow m = -2, \frac{1}{2}$

69. B

Sol. $37^{2024} + 43^{2024} = 16\lambda + 2$

70. C

Sol. $13 + (N-1)\,20 \le 8093$

$\Rightarrow N \le 405$

71. C

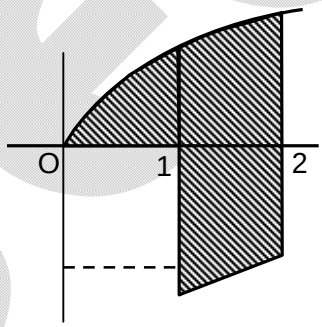
Sol. $a + b = 5$

72. C

Sol. $a = 7$

73. C

Sol. Required area $= \int_0^2 4\sqrt{x}\,dx + \frac{1}{2}(2023 + 2022)$



74. D

Sol. $\sqrt{2}\sum_{k=1}^{9}\left[\tan\left(\theta + (2k+1)\frac{\pi}{8}\right) - \tan\left(\theta + (2k-1)\frac{\pi}{8}\right)\right]$

$\Rightarrow \sqrt{2}\left(\tan\left(\theta + \frac{19\pi}{8}\right) - \tan\left(\theta + \frac{\pi}{8}\right)\right) = \left(2 + \frac{2}{\sqrt{3}}\right)\sqrt{2}$

$\Rightarrow \frac{1}{\cos\left(\theta + \frac{\pi}{8}\right)\cos\left(\theta + \frac{3\pi}{8}\right)} = 2\left(1 + \frac{1}{\sqrt{3}}\right)\sqrt{2}$

$\Rightarrow \cos\frac{\pi}{4} + \cos\left(2\theta + \frac{\pi}{2}\right) = \frac{\sqrt{3}}{(\sqrt{3}+1)\sqrt{2}}$

$\Rightarrow \sin 2\theta = \frac{\sqrt{3}-1}{2\sqrt{2}} \Rightarrow \theta = \frac{\pi}{24}, \frac{11\pi}{24}$

75. C

Sol. $x > 0,\ \tan^{-1}\left(\frac{10x}{25 - x^2}\right) = \frac{\pi}{6} \Rightarrow x = 10 - 5\sqrt{3}$

$x < 0,\ \tan^{-1}\left(\frac{10x}{25 - x^2}\right) = -\frac{\pi}{3} \Rightarrow x = -\frac{5}{\sqrt{3}}$

76. C

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Sol. $a = 5, \angle B = \frac{5\pi}{12}$

$\sin A = \frac{a}{2R} = \frac{5}{5\sqrt{2}} = \frac{1}{\sqrt{2}} \Rightarrow A = \frac{\pi}{4}$

Also $b = 2R \sin B = 5\sqrt{2}\left(\frac{\sqrt{3}+1}{2\sqrt{3}}\right) = \frac{5(\sqrt{3}+1)}{2}$

$\angle C = \frac{\pi}{3}$

Hence $\Delta = \frac{1}{2}ab\sin C = \frac{1}{2}\times 5\times 5\frac{(\sqrt{3}+1)}{2}\times\frac{\sqrt{3}}{2} = \frac{25\sqrt{3}(\sqrt{3}+1)}{8}$

77. B

Sol. $32\sin^5\alpha + 243\cos^5\beta + 1 + 1 + 1 \geq 5(2^5 3^5 \sin^5\alpha \cos^5\beta)^{1/5} \geq 30\sin\alpha\cos\beta$

$\Rightarrow \quad 32\sin^5\alpha = 243\cos^5\beta = 1$

$\Rightarrow \quad \sin\alpha = \frac{1}{2}, \cos\beta = \frac{1}{3}$

$\cos 2\alpha = \frac{1}{2}, \cos 2\beta = -\frac{7}{9}$

78. C

Sol. $(a, a) \in R$: R is reflexive

Let $a = \frac{q}{p}$ and b is rational then $q - pb + q$ is irrational but $pb - p\frac{q}{p} + q$ is rational

$\therefore$ R is not symmetric

Let $pa - pb + q$ and $pb - pc + q$ both are irrational but $p(a - c) + 2q$ may be rational if $a - c = -\frac{2q}{p}$

$\therefore$ non transitive.

79. C

Sol. $\int \frac{x\cos x - \sin x}{x\sqrt{x^2 - \sin^2 x}}dx = \int \frac{\frac{x\cos x - \sin x}{\sin^2 x}}{\frac{x}{\sin x}\sqrt{\left(\frac{x}{\sin x}\right)^2 - 1}}dx = \operatorname{cosec}^{-1}\left(\frac{x}{\sin x}\right) + c$

80. A

Sol. $\frac{1}{(y+2024)^2}\frac{dy}{dx} + \frac{1}{y+2024}x^2 = e^{x^3/3}$

$\Rightarrow \quad \frac{dt}{dx} - tx^2 = e^{x^3/3}; t = -\frac{1}{y+2024}$

$-\frac{1}{y+2024}e^{-x^3/3} = x + c$

$\because \ y(0) = -\frac{6071}{3} \ \therefore \ c = -3$

$y + 2024 = -\frac{1}{(x-3)e^{x^3/3}}$

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SECTION – B

81. 1728

Sol. $n(C) = (2^{506} - 1)2^{1518}$

$m - p = 1012.$

82. 16

Sol. Equation of perpendicular bisector of the line segment joining the points (3, 5) and (7, 9) is $x + y - 12 = 0$. Solving it with the equation of diameter $3x - y - 12 = 0$ we get centre as (6, 6)

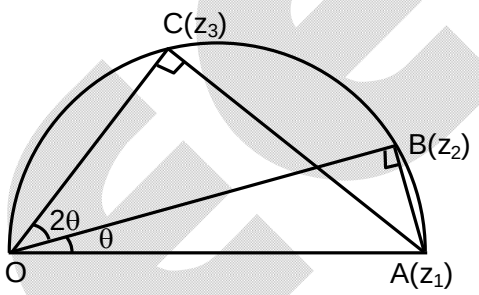
$\therefore$ Area of rectangle = $2\sqrt{2} \times 4\sqrt{2} = 16$

83. 6

Sol. $\frac{z_3}{z_1} = \cos 3\theta e^{i3\theta}$

$\frac{z_2}{z_1} = \cos\theta e^{i\theta}$

$\left(\frac{z_2}{z_1}\right)^3 = \frac{\cos^3\theta}{\cos 3\theta} \times \frac{z_3}{z_1} \Rightarrow \frac{\cos^3\theta}{\cos 3\theta} = \frac{5+3\sqrt{3}}{8} \Rightarrow \theta = \frac{\pi}{12}$



84. 171

Sol. Let $P_n = \alpha^n + \beta^n$ then $P_{n+1} - P_{n-1} = 13P_n$

85. 300

Sol. 111138, 111146, 113222, 111226, 111432

$\frac{6!}{4!} \times 2 + \frac{6!}{3!2!} \times 2 + \frac{6!}{3!}$

$= 60 + 120 + 120 = 300$

86. 14

Sol. $5^{n(n-1)} 3^{(n-1)^3} = 5^{42} 3^{216} \Rightarrow n = 7$

$$\pi \int_{n-14}^{n} x \sin \pi x \, dx = 2\pi \left[\left(-\frac{x \cos \pi x}{\pi} \right)_0^7 + \frac{1}{\pi} \int_0^7 \cos \pi x \, dx \right]$$

87. 280

Sol. $L_1 : \frac{x-8}{8} = \frac{y-0}{-6} = \frac{z-0}{0}$

$L_2 : \frac{x}{0} = \frac{y+8}{8} = \frac{z}{-5}$

$$d = \frac{\begin{vmatrix} 8 & 8 & 0 \\ 8 & -6 & 0 \\ 0 & 8 & -5 \end{vmatrix}}{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 8 & -6 & 0 \\ 0 & 8 & -5 \end{vmatrix}} = \frac{280}{\sqrt{1649}}$$

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88. 10118

Sol. $(2023\vec{a}+2024\vec{b})\cdot\left\{\left(2|\vec{b}|^2\,\vec{a}+3|\vec{a}|^2\,\vec{b}\right)\right\}$

$=(2023\vec{a}+2024\vec{b})\cdot(2\vec{a}+3\vec{b})$

89. 180

Sol. Possible values of f(1), f(2), f(3) are

(1, 2, 3), (2, 3, 4), (3, 4, 5), (4, 5, 6), (5, 6, 7)

(1, 3, 5), (2, 4, 6), (3, 5, 7), (1, 4, 7)

$P(E)=\frac{n(E)}{n(S)}=\frac{18.4!}{7!}=\frac{18.4!}{7.6.5.4!}=\frac{3}{35}$

90. 200

Sol. $x + y = 21$

$x^2 + y^2 = 225$

$\Rightarrow x = 9, y = 12$

$\therefore$ M. D $=\frac{\sum_{i=1}^{6}|x_i-\bar{x}|}{6}=\frac{10}{3}$

FIITJEE